

Chemistry

Middletown High School | Grade 12, Fall Semester 2026

Teacher: Dr. P. Nandakumar
Room: 112 Lab
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Course Description

A laboratory based chemistry course covering atomic structure, bonding, the mole and reactions. Students learn to predict what a reaction will do and then measure whether it did.

Course Units

Weeks 1 to 3	Unit 1: Atomic Structure and Periodicity Protons, neutrons and electrons, electron configuration, and periodic trends in atomic radius, ionization energy and electronegativity across a period and down a group.
Weeks 4 to 6	Unit 2: Chemical Bonding Ionic, covalent and metallic bonding, Lewis structures, molecular shapes and polarity.
Weeks 7 to 10	Unit 3: Stoichiometry and the Mole The mole is 6.022×10^{23} particles. Moles = mass divided by molar mass. Balance the equation before using mole ratios. The limiting reagent sets the maximum product.
Weeks 11 to 13	Unit 4: Acids, Bases and pH Strong and weak acids and bases, the pH scale, neutralization and titration technique.
Weeks 14 to 16	Unit 5: Organic Chemistry Foundations Hydrocarbons, functional groups and naming simple organic compounds.

Key Dates This Semester

Sep 30	Unit 2 Bonding Quiz
Oct 21	Mid-Term Exam, covers Units 1 to 3
Nov 18	Titration Lab Report due
Dec 14	Final Exam, all units

School wide dates

Sep 8	First day of classes
Oct 19 to Oct 23	Mid-term examination week
Nov 6	Parent teacher conferences
Nov 25 to Nov 27	Thanksgiving break, no classes
Dec 14 to Dec 18	Final examination week

Materials

Scientific calculator, lab notebook, closed toe shoes for lab days, periodic table (provided).

Grading Policy

25 percent	Laboratory work
10 percent	Homework
15 percent	Quizzes
30 percent	Unit tests
20 percent	Final exam

A = 90 to 100 percent, B = 80 to 89, C = 70 to 79, D = 60 to 69, F = 59 and below.

Late and Make-Up Work

Work handed in late without an excused absence receives a maximum of 50 percent credit. With an excused absence, work is due within one week of returning. It is the student's responsibility to ask what was missed. Make-up tests are held on Thursdays after school and must be booked at least one day ahead.

Academic Honesty

All work submitted must be your own. Copying another student's work, or submitting text or code generated by someone or something else as your own, is treated as academic dishonesty and results in a zero for that assignment and a call home.

Extra Help

Extra help is available Tuesdays and Thursdays, 3:15 to 4:00 pm, in the room listed above.